Development and Performance Evaluation of a Pedal Operated Seed Cleaner (POS-Cleaner)

Abstract

Traditional grain cleaning methods are labor-intensive, time-consuming, and inefficient. Mechanical cleaners are expensive. A pedal-operated seed cleaner (PoS-Cleaner) was developed using engineering principles. It consists of a pedaling system, hopper, centrifugal fan, and adjustable sieves.

Cleaning rates:

- Maize: 576.5 kg/h
- Beans: 375.8 kg/h
- Groundnuts: 377.4 kg/h

Highest efficiencies:

- Maize: 95.09%
- Beans: 87.61%
- Groundnuts: 81.67%

1. Introduction

Grains are a major food source worldwide, providing carbohydrates, proteins, and nutrients. Uganda faces post-harvest losses due to inefficient cleaning. Manual methods are slow and inconsistent. Mechanical cleaners are expensive and limited.

This study develops a multipurpose pedal-operated cleaner for smallholder farmers.

2. Materials and Methods

2.1 Machine Description

The PoS-Cleaner includes:

- Pedaling system
- Hopper
- Centrifugal fan
- Three sieves (adjustable)

Operation:

- Seeds enter hopper
- Air removes light materials
- Sieves separate by size

- Clean seeds collected

2.2 Design Components

Includes:

- Hopper design
- Rotating sieves
- Belt and pulley system
- Chain drive
- Shaft and blower

Engineering equations were used to calculate:

- Volume
- Speed
- Power
- Efficiency

2.3 Performance Evaluation

Tested using maize, beans, and groundnuts.

Parameters measured:

- Separation loss (SL)
- Separation efficiency (SE)
- Cleaning loss (CL)
- Cleaning efficiency (CE)
- Cleaning rate (CR)
- Seed damage (SD)

3. Results and Discussion

3.1 Performance

- High separation efficiency
- Low cleaning loss
- High cleaning rates
- Seed damage <1%

3.2 Cleaning Rate

- Faster than traditional methods (3–5x improvement)

3.3 Optimization

Optimal sieve sizes:

- Maize: 13 mm
- Beans: 16 mm
- Groundnuts: 10 mm

4. Conclusions

The PoS-Cleaner improves efficiency, reduces labor, and is suitable for rural farmers without electricity.

It provides:

- Faster cleaning
- Better quality seeds
- Increased productivity

References

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